# 2024 JC2 H2 GEOGRAPHY PRELIMINARY EXAMINATION
# PAPER 1 *Suggested Answers*

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***1. Resource 1 shows the location and amount of economically viable oil reserves at different oil prices in 2016. Resource 2 shows information produced by the Canadian government about the impacts of tar sand developments. Resource 3 shows a tar sand oil production area in Alberta, Canada. Resource 4 shows minerals and the associated geology found in part of northwest England.***

***1a) Citing data from Resource 1, describe the relationship between oil prices and the location and amount of economically viable oil reserves shown in Resource 1. [4]***

*Award 1 mark for an accurate description of the relationship.*
*Award 1 mark for each citing and description of data to support the identified relationship, to a maximum of 3 marks.*
*Award a maximum of 1 additional mark for further development of each description, where applicable.*

Possible responses include:
As the price of oil per barrel increases the amount of viable oil also increases in tandem, as does the number of countries with viable oil.
For example, at US$20 there is about 350Gb of viable oil reserves.
At US$100, there is nearly four times the amount of viable oil reserves (≈1290Gb).

***1b) Explain the differences between proven reserves, conditional reserves, and speculative reserves. [4]***

*Award 1 mark for each explanation of the differences between the three types of reserves, to a maximum of 4 marks.*
*Award a maximum of 1 additional mark for further development of each explanation, where applicable.*

Possible responses include:
Proven reserves are mineral deposits that have been discovered and are known to be economically extractable under current demand, price and technological conditions.
Conditional reserves refer to mineral deposits that have already been discovered but are not economic to work at prevailing price levels.
The extent to which a natural resource is considered as a proven resource or a conditional resource is dynamic, bi-directional, and is dependent upon a host of interlinked factors.
Speculative resources commonly refer to those resources that might be found in unexplored areas of the earth that are thought to have favourable geological conditions.

***1c) With reference to Resources 2 and 3, explain how the expansion of tar sand developments can impact Canada's progress towards sustainable development. [7]***

*Award 1 mark for each relevant explanation, to a maximum of 7 marks.*
*Award a maximum of 1 additional mark for further development of each explanation, where applicable.*

Possible responses include:
Sustainable development refers to development that meets the needs of the present without compromising the ability of future generations to meet their own needs.

An expansion of tar sand development can potentially increase Canada's oil exports and improve the country's balance of payments and energy security.
However, the extraction of tar sands can present serious environmental problems.
Also, their extraction consumes vast quantities of energy and water.

Resource 2 shows that the tar sand mines have already swallowed an area larger than New York City. This will affect the traditional way of life of the Indigenous community of Fort McKay.
Thus, the expansion of tar sand developments can also negatively impact Canada's pursuit of sustainable development in the social aspect.

## *1d) Explain the indicators that can be used to measure the level of economic growth and the degree of social inclusion in a country respectively. [4]*

*Award 1 mark for each suggestion of a relevant indicator to measure economic growth and social inclusion respectively, to a maximum of 2 marks.*
*Award 1 mark for each explanation to a maximum of 2 marks.*
*Award a maximum of 1 additional mark for further development of each description, where applicable.*

Possible responses include:
Gross national income (GNI) is a useful indicator to measure the level of economic growth in a country as such statistical information have been collected for a long period of time in many countries. This is especially so for the developed countries where the majority of their economic transactions, production of goods and services and labour occur formally.

One indicator that can be used to measure the degree of social inclusion is that of adult literacy, especially that of female adult literacy. This is because two-thirds of the world's adult illiterates are women.

## *1e) Cite data from Resource 4 to describe the relationship(s) between mineral mining sites and rock types. [4]*

*Award 1 mark for each description of a relevant relationship, to a maximum of 2 marks.*
*Award 1 mark for each data cited to support the identified relationship(s) to a maximum of 2 marks.*
*Award a maximum of 1 additional mark for further development of each description, where applicable.*

Possible responses include:
Resource 4 shows a clear association between rock type and mineral mining sites.
It is observed that mineral mining sites occur much more frequently in metamorphic and igneous rocks.
There are only 5 mineral mining sites in sedimentary rocks. In contrast, at least 15 mineral mining sites can be found in Borrowdale and Eycott Volcanics alone.
Skiddaw Group (mainly metamorphic rocks) have the most variety of mineral mining sites. Graphite is the only mineral not found in the Skiddaw Group.
The sedimentary rocks only have 3 different minerals found in them - Baryte, Haematite, Lead and zinc.

## *1f) Explain the characteristics of extractive industries such as that of mineral mining. [7]*

*Award 1 mark for each explanation of the characteristics of the extractive industries, to a maximum of 7 marks.*
*Award a maximum of 1 additional mark for further development of each explanation, where applicable.*

Possible responses include:
The extractive industry is highly capital and technologically intensive.
Much investment and expenditure are required in the various operations of this industry including that of exploration, extraction, processing, and even transportation and logistical support.
Another characteristic of the extractive industry is that they tend to be sited in far-flung locations.
Such geographical remoteness often encourages the development of enclave tendencies.
The high-risk nature of business is another key characteristic of the extractive industry.
There are many risks involved in the exploration of new supplies.

***2. Resource 5 shows the temporal distribution of reported mass movement events in Western Cape, South Africa. Resource 6A shows the percentage of land that is very highly susceptible to landslides in the districts of Kerala, India. Resource 6B shows the districts of Kerala, India. Resource 7 shows an abstract from a news article on the causes of the 30 July 2024 landslide near Meppadi in Wayanad district, Kerala. Resource 8 shows the land use map of the Mekong Delta in 2015.***

### *2a) Citing data from Resource 5, compare the temporal distribution of flows with the temporal distribution of falls. [4]*

*Award 1 mark for each comparison of the temporal distribution of flows with the temporal distribution of falls, to a maximum of 4 marks. Award a maximum of 1 additional mark for further development of each comparison, where applicable.*
*Award a maximum of 3 marks for responses which do not describe both similarities and differences.*

Possible responses include:
Both flows and falls have the highest number of reported events in July.
There are no reported events of both flows and falls in the months of February, March, May and October.
January, September, November and December have reported events of falls but not flows.

### *2b) Distinguish the processes of mass movement, water erosion and weathering. [4]*

*Award 1 mark for each description of the processes of mass movement, water erosion and weathering, to a maximum of 3 marks.*
*Award a maximum of 1 additional mark for further development of each description, where applicable.*

Possible responses include:
Mass movement is the downslope movement of earth materials under the influence of gravity.
It occurs when shear stress exceeds shear strength, i.e. the safety factor being less than one.
On the other hand, water erosion refers to removal of soil by water and transportation of the eroded materials away from the point of removal.
Weathering refers to the physical and chemical breakdown of rocks and minerals in situ.
Weathering processes may thus be classified as physical, chemical or biological.

### *2c) Citing data from Resources 6A and 6B, describe the spatial distribution of land that is very highly susceptible to landslides in the districts of Kerala, India. [4]*

Award 1 mark for each description of the spatial distribution of land that is very highly susceptible to landslides in the districts of Kerala, India, to a maximum of 4 marks.
Award a maximum of 1 additional mark for further development of each description, where applicable.

Possible responses include:
The northern districts have a higher percentage of land that is very susceptible to landslides as compared to the central and southern districts.
The northern district of Wayanad, has the highest percentage of land that is very susceptible to landslides at 31.54%.
On the other hand, the southern district of Alappuzha has the lowest percentage of land that is very susceptible to landslides at 0%.
An anomaly is the southern district of Idukki, which has 31.08% of land that is very susceptible to landslides. This figure is much higher as compared to its neighbouring districts.

## *2d) With reference to Resource 7, suggest reasons for the occurrence of the 30 July landslide near Meppadi in Wayanad district. [6]*

*Award 1 mark for each explanation of the reasons for the occurrence of the 30 July landslide near Meppadi in Wayanad district, to a maximum of 6 marks.*
*Award a maximum of 1 additional mark for further development of each explanation, where applicable.*

Possible responses include:
One reason could be heavy precipitation.
This is because with heavy precipitation, the high moisture content in the soils will load the slopes, increasing the weight exerted. The landslide may then occur when shear stress exceeds shear strength.
From Resource 7, heavy precipitation is a plausible reason as it indicates that Wayanad was subjected to unrelenting monsoon rain with 28 cm of rain received in the 24 hours before the landslides struck.

Another reason could be human activities such as road-building.
When roads are constructed in hilly regions, the construction would necessitate cutting into the sides of the hills. This action further steepens the slopes locally which increases shear stress since slope materials are now more susceptible to the pull of gravity.
This reason could apply to Wayanad district in Kerala as Resource 7 states that as India has sought to develop economically over the past few decades, it has built roads, bridges and mines in hilly regions. In Kerala's context, the latter refers to the Western Ghats mountain range to the east.

## *2e) Explain the role of geomorphic processes in the formation of deltas. [8]*

*Award 1 mark for each explanation of the role of geomorphic processes in the formation of deltas, to a maximum of 8 marks.*
*Award a maximum of 1 additional mark for further development of each explanation, where applicable.*

Possible responses include:
Deltas are formed through geomorphic processes carried out by rivers, waves and tides.
The river transports eroded sediments downstream. As the river nears its terminus, its velocity decreases due to the flattening of the landscape and the influence of gravitational forces. Consequently, it starts depositing this sediment as it enters the calmer waters of the coastal area.
As sediment accumulates, it begins to block the main river channel. This causes the river to split into smaller channels known as distributaries. The sediment piles up, gradually building the delta.
As waves approach the shoreline, they carry and deposit sediments picked up from offshore areas. The fine-grained sediments such as sand, silt and clay are deposited near the river mouth.
These waves also contribute to the formation of bars and shoals, which help in the deposition process, in the nearshore environment. These features help to dissipate wave energy, slow down the water flow and create conditions conducive to sediment deposition which facilitates the buildup of material that contributes to delta growth.
Tidal currents can erode and transport sediment. During high tides, tidal-laden water from the river may be transported and deposited in the delta region. During low tides, sediment may be transported seaward.

## *2f) With references to Resource 8, explain the ecosystem services provided by deltas. [4]*

*Award 1 mark for each explanation of the ecosystem services provided by deltas, to a maximum of 4 marks.*
*Award a maximum of 1 additional mark for further development of each explanation, where applicable.*

Possible responses include:
One ecosystem service provided by deltas could be provisioning services in the form of the provision of food through agricultural activities that can be conducted in deltas. This is seen in Resource 8 where double rice crops and triple rice crops take up 712 722 ha and 957 462 ha of land respectively.
Another ecosystem service could be regulating services. These could be in the form of carbon storage and climate regulation as vegetation in the deltas take in carbon dioxide and produce oxygen during photosynthesis.

### *3. 'The world's resource base will always be enough to support population growth.' Evaluate this statement. [20]*

While the neo-conservationist school of thought takes on a cautionary position regarding the potential availability of the world's resource base to always support continuous population growth, proponents of the hedonist school of thought that believes in the potential of technology and human innovation have a more optimistic stance and tend to agree that the resource base will continue to be enough.

- This essay considers the merits and limitations of both stances to contend that the issue of whether there are enough resources depends on more than just technological inventiveness but also factors like governance and accessibility.

**Possible Points of Discussion:**

- Ester Boserup's views on role of technology in expanding resource base
- Limitations to Ester Boserup's views: Boserup's lack of consideration of degradation of the resource base because of technological innovations; Boserup's lack of consideration of unevenness in access to resources across the population
- Thomas Malthus' views on limit imposed by the natural environment on population numbers

### *4. 'The environmental impacts of the GPNs of TNCs are the most damaging in host economies.' Evaluate this statement. [20]*

A TNC is a firm that controls and coordinates other firms' activity in more than one country.

- This control and coordination is made possible either because a TNC owns its subsidiaries abroad or because it is the single largest buyer of outputs from these foreign firms.
- In fact, the control and coordination practised by TNCs is reflected in their global production networks (GPNs). GPNs can be understood as meshes of intra- and inter-firm relationships, such as with suppliers, sub-contractors and strategic partners, that lie behind nearly all the products and services that we consume on a daily basis.
- It is the aim of this essay to assert that it may be an overgeneralisation to consider that the GPNs of all TNCs will have damaging environmental impacts in host economies.

**Possible Points of Discussion:**

- Case study of Shell's operations in Ogoniland, Nigeria – evaluate the environmental impact the GPNs of the TNCs have had.
- Case study of Toyota's adoption of the Toyota Production System to eliminate waste in the USA.

### *5. Evaluate the variations in the effects of fluvial floods on places in the humid tropics. [20]*

Fluvial flooding occurs when the discharge of a river exceeds its bankfull capacity.

- It is the aim of this essay to evaluate the variations in the socio-economic and environmental effects of fluvial floods on places in the humid tropics.

**Possible Points of Discussion:**

- Floods can have significant negative effects on the economic growth of the affected region, including damage and destruction to property and infrastructure.
- In addition to the damage costs, fluvial flooding can result in lost productivity, reduced hours worked and losses in GDP for the affected region
- Flooding can also cause major disruptions to the provision of services and utilities, and lead to many other secondary hazards and problems that affect the liveability of the environment.

- Fluvial floods can result in a loss of wildlife and biodiversity in the flooded region, creating long-term impacts for the surviving wildlife.
- While fluvial floods potentially result in much socio-economic and environmental impacts, it must be noted that these negative effects vary across places due to the different degrees of preparedness and investment made in flood mitigation strategies.

## *6. 'Climate alone dictates the role of geomorphic processes in the formation of karst landforms.' Evaluate this statement. [20]*

Karst landforms are distinctive landform features developed in limestone bedrock.

- They exhibit a variety of physical characteristics, but the most distinctive landforms associated with karst landscapes in the humid tropics include cockpits, cones and tower karst.
- Given that chemical weathering of limestone is the key process behind Karst landform formation, it is reasonable to say that climate is an important factor to karst formation as the climatic controls of temperature and precipitation directly affect the rate and intensity of such weathering processes.
- However, it is erroneous to claim that climate alone dictates the geomorphic processes behind karst development.
- This essay contends that climate works in tandem with other factors such as the unique geological characteristic of limestone rock as well as topography to account for the development of karst landforms.

**Possible Points of Discussion:**

- Explain the key role climate plays in the development of karst landscapes - large-scale karst landscapes are best developed in a hot and humid climate; reference to Van't Hoff's Law and Peltier's model
- Explain the significance of geology and topography in karst development
- Explain the role of other factors such as tectonics and time in the development of karst landscapes.

***Levels marked:***

| Levels | Marks | Generic Level Descriptors for H2 Essays |
|---|---|---|
| **5** | 18–20 | Evaluation is consistently analytical and coherent. Response is well-supported by relevant material, including the effective use of examples. Response features accurate geographical knowledge and reflects good understanding of the subject content relevant to the question. |
| **4** | 14–17 | Evaluation is analytical and coherent. Response is mostly well-supported by relevant material, including the appropriate use of examples. Response features accurate geographical knowledge and reflects adequate understanding of the subject content relevant to the question. |
| **3** | 10–13 | Evaluation is broadly analytical and generally coherent. Response is moderately well-supported by relevant material, including some appropriate use of examples. Response features accurate geographical knowledge and reflects adequate understanding of the subject content relevant to the question. |
| **2** | 6–9 | Response is largely descriptive with limited analysis and evaluation. Response is partly coherent and may lack clarity in parts. Response is poorly supported by relevant materials, including the limited use of examples. Response features inaccurate geographical knowledge and poor understanding of the subject content relevant to the question. |
| **1** | 1–5 | Response is descriptive with no analysis or evaluation. Response is fragmented and lacks clarity. Response consists of unsupported assertions. Response features largely inaccurate geographical knowledge and a lack of understanding of the subject content relevant to the question. |
| **0** | 0 | No creditworthy response |